Arithmetic & Number Sense

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

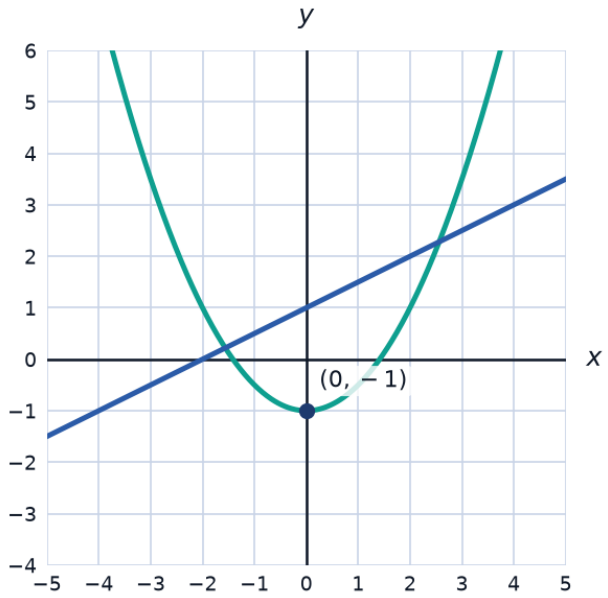
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Order of operations

Evaluate $12 + 6 \times (9 - 5) \div 3$.

2 [Foundational] • Target: Order of operations

Evaluate $8 + 2 \times 5^2 - 30 \div 6$.



3 [Foundational] • Target: Order of operations

Evaluate $\frac{18 - 3 \times 4}{2} + 7$.

4 [Foundational] • Target: Order of operations

Evaluate $(7 - 2)^2 - 3 \times (6 - 4)$.

5 [Foundational] • Target: Powers and roots

Evaluate 3^4.

6 [Foundational] • Target: Powers and roots

Evaluate $\sqrt{144}$.

7 [Foundational] • Target: Powers and roots

Evaluate $2^3 \times 2^2$ using the laws of exponents.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Powers and roots

Evaluate $\sqrt{81} + \sqrt{16}$.

9 [Intermediate] • Target: Working with fractions

Compute $\frac{3}{4} + \frac{5}{6}$.

10 [Intermediate] • Target: Working with fractions

Compute $\frac{7}{8} - \frac{1}{3}$.

11 [Intermediate] • Target: Working with fractions

Compute $\frac{2}{3} \times \frac{9}{10}$.

12 [Intermediate] • Target: Working with fractions

Compute $\frac{5}{6} \div \frac{5}{9}$.

13 [Intermediate] • Target: Working with fractions

Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?

14 [Intermediate] • Target: Decimals

Compute $3.45 + 2.7$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Decimals • Cross-Lesson: 3.01 Algebra Essentials

Compute $8.6 - 3.25$.

16 [Challenging] • Target: Decimals • Cross-Lesson: 4.01 Geometry & Measurement

Compute 1.2×0.5.

17 [Challenging] • Target: Decimals • Cross-Lesson: 5.01 Quantitative Comparisons

Compute $7.5 \div 0.25$.

18 [Challenging] • Target: Place value and order of magnitude • Cross-Lesson: 6.01 Sequences & Patterns

Write $4{,}500{,}000$ in the form $a \times 10^n$ with $1 \le a < 10$.

19 [Challenging] • Target: Place value and order of magnitude • Cross-Lesson: 7.01 Data Analysis & Probability

Which is greater, 6×10^4 or $58{,}000$?

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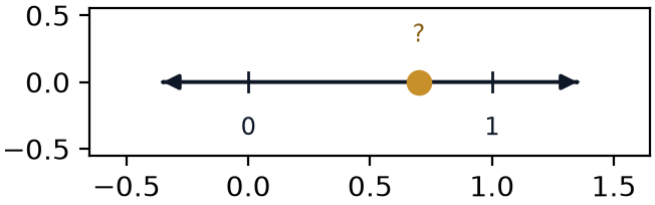
20 [Challenging] • Target: Place value and order of magnitude • Cross-Lesson: 8.01 Word Problems & Reasoning

Order from least to greatest: 0.6, $\frac{2}{3}$, 0.65.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Comparing fractions and decimals

Which value is largest: the point shown on the number line, $\frac{3}{5}$, or 0.55?



a. The marked point (0.7)

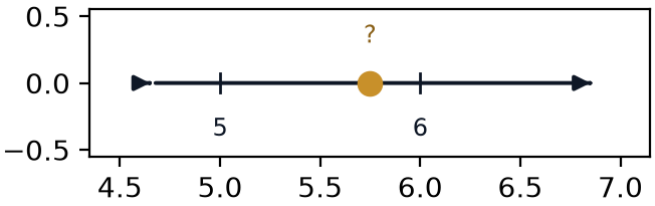
b. $\frac{3}{5}$

c. They are all equal

d. 0.55

22 [Foundational] • Target: MCQ — Decimal operations

The number line marks the value of $9.2 - 3.45$. Where does it land?



a. 5.75

b. 6.25

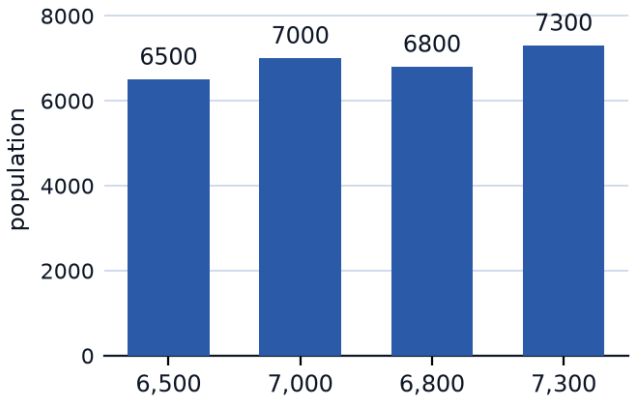
c. 5.85

d. 5.65

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23 [Foundational] • Target: MCQ — Scientific notation & place value

The bar chart shows four city populations. Which one equals 7×10^3?



a. 7,000

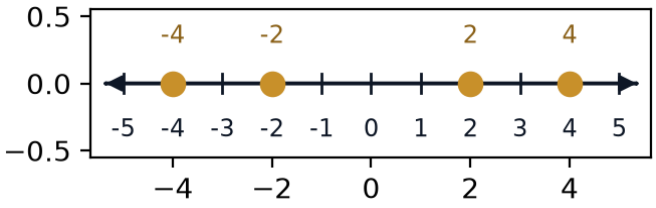
b. 6,800

c. 6,500

d. 7,300

24 [Foundational] • Target: MCQ — Absolute value & signed numbers

Two of the four points on the number line represent a pair of numbers that are equal in absolute value but opposite in sign. Which pair?



a. −2 and 2

b. −4 and 4

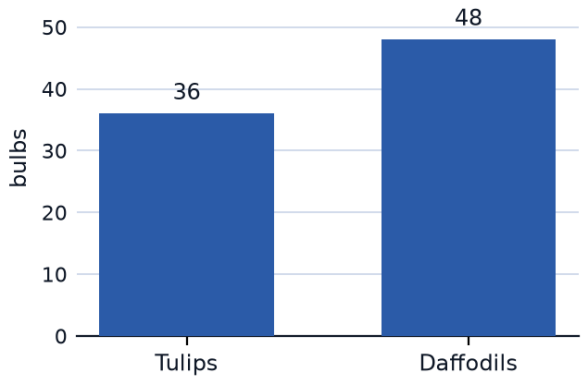
c. −2 and 4

d. −4 and 2

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25 [Foundational] • Target: MCQ — GCD, LCM & factors

The bar chart shows a gardener's bulb counts. They want to plant them in identical rows with the largest possible number of bulbs per row, each row containing only one type of flower. How many bulbs per row?



a. 4

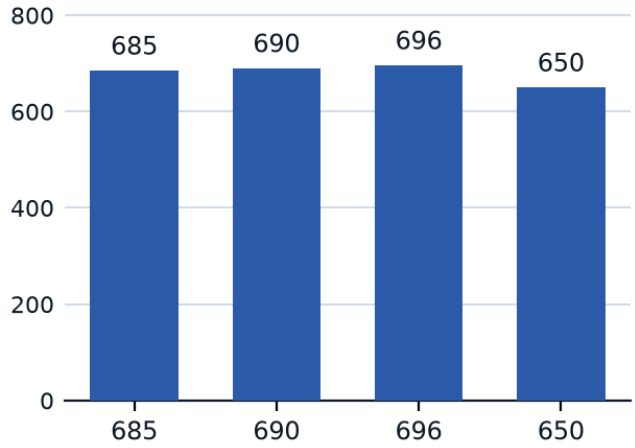
b. 24

c. 6

d. 12

26 [Foundational] • Target: MCQ — Divisibility rules

The bar chart shows four candidate numbers. Which one is divisible by both 5 and 6?



a. 690

b. 650

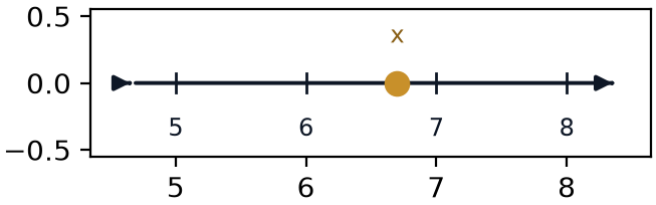
c. 685

d. 696

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27 [Foundational] • Target: MCQ — Estimation & rounding

The number line shows the value x rounded to the nearest whole number. If $x = 6.7$, where does the rounded value land?



a. 6

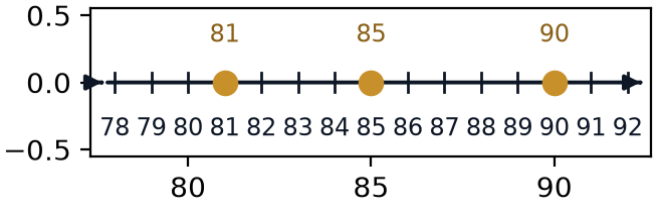
b. 6.5

c. 8

d. 7

28 [Foundational] • Target: MCQ — Number properties

The number line marks 85 between consecutive multiples of 9. Find the remainder when 85 is divided by 9.



a. 3

b. 4

c. 5

d. 76

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Order of operations

Evaluate $20 - 4 \times 3 + 6 \div 2$.

a. 11

b. 27

c. 9

d. 17

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30 [Intermediate] • Target: MCQ — Exponents and roots

Evaluate 2^5.

a. 16

b. 25

c. 32

d. 10

31 [Intermediate] • Target: MCQ — Fraction operations

Compute $\frac{2}{5} + \frac{1}{4}$.

a. $\frac{1}{2}$

b. $\frac{11}{20}$

c. $\frac{13}{20}$

d. $\frac{1}{3}$

32 [Intermediate] • Target: MCQ — Comparing fractions and decimals

Order from least to greatest: 0.62, $\frac{5}{8}$, 0.6.

a. $0.62 < 0.6 < \frac{5}{8}$

b. $0.6 < \frac{5}{8} < 0.62$

c. $\frac{5}{8} < 0.6 < 0.62$

d. $0.6 < 0.62 < \frac{5}{8}$

33 [Intermediate] • Target: MCQ — Scientific notation & place value

Evaluate 3.4×10^4 as an ordinary number.

a. 3,400,000

b. 3,400

c. 340,000

d. 34,000

34 [Intermediate] • Target: MCQ — Absolute value & signed numbers

Evaluate $|-9| + |-4|$.

a. 5

b. -13

c. 13

d. 36

35 [Intermediate] • Target: MCQ — GCD, LCM & factors

How many factors does 60 have?

a. 6

b. 8

c. 12

d. 10

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36 [Intermediate] • Target: MCQ — Estimation & rounding

Estimate 312×48 by rounding each factor to the nearest ten, then compare to the exact product $14{,}976$.

a. Estimate $310 \times 50 = 15{,}500$, far from the exact value

b. Estimate $300 \times 50 = 15{,}000$, close to the exact value

c. Estimate $320 \times 40 = 12{,}800$, close to the exact value

d. Estimate $310 \times 50 = 15{,}500$, close to the exact value

37 [Intermediate] • Target: MCQ — Decimal operations

Evaluate $(8 + 7)^2 - 2 \times 7$.

a. 211

b. 220

c. 207

d. 215

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Scientific notation & place value • Cross-Lesson: 6.01 Sequences & Patterns

Evaluate $34 - 5 \times 6 + 18 \div 6$.

a. 16

b. 11

c. 3

d. 7

39 [Challenging] • Target: MCQ — Mixed numbers & improper fractions • Cross-Lesson: 7.01 Data Analysis & Probability

Evaluate $36 + 5 \times (5 - 1)$.

a. 56

b. 60

c. 65

d. 52

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40 [Challenging] • Target: MCQ — Absolute value & signed numbers • Cross-Lesson: 8.01 Word Problems & Reasoning

Compute $\frac{6}{8} - \frac{1}{8}$.

a. $\frac{3}{4}$

b. $\frac{3}{8}$

c. $-\frac{5}{8}$

d. $\frac{5}{8}$

41 [Challenging] • Target: MCQ — GCD, LCM & factors • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Compute $\frac{5}{6} - \frac{3}{5}$.

a. $\frac{7}{30}$

b. $\frac{1}{6}$

c. $-\frac{7}{30}$

d. $\frac{4}{15}$

42 [Challenging] • Target: MCQ — Divisibility rules • Cross-Lesson: 3.01 Algebra Essentials

Find the greatest common divisor (GCD) of 90 and 72.

a. 36

b. 18

c. 21

d. 15

43 [Challenging] • Target: MCQ — Estimation & rounding • Cross-Lesson: 4.01 Geometry & Measurement

Find the least common multiple (LCM) of 19 and 16.

a. 310

b. 304

c. 305

d. 303

44 [Challenging] • Target: MCQ — Number properties • Cross-Lesson: 5.01 Quantitative Comparisons

A shipment of 310 books is divided as equally as possible among 15 shelves. How many books does each shelf receive, and how many are left over?

a. $20, 11$

b. $21, 10$

c. $10, 20$

d. $20, 10$

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45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Sequences & Patterns

Convert $\frac{29}{6}$ to a mixed number.

a. $4\frac{5}{6}$

b. $5\frac{5}{6}$

c. $5\frac{4}{5}$

d. $3\frac{5}{6}$

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